

Recovering Deleted and Damaged Files (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

Student:

Prakriti Dhungel

Email:

prakriti.dhungel98@gmail.com

Time on Task:

11 hours, 4 minutes

Progress:

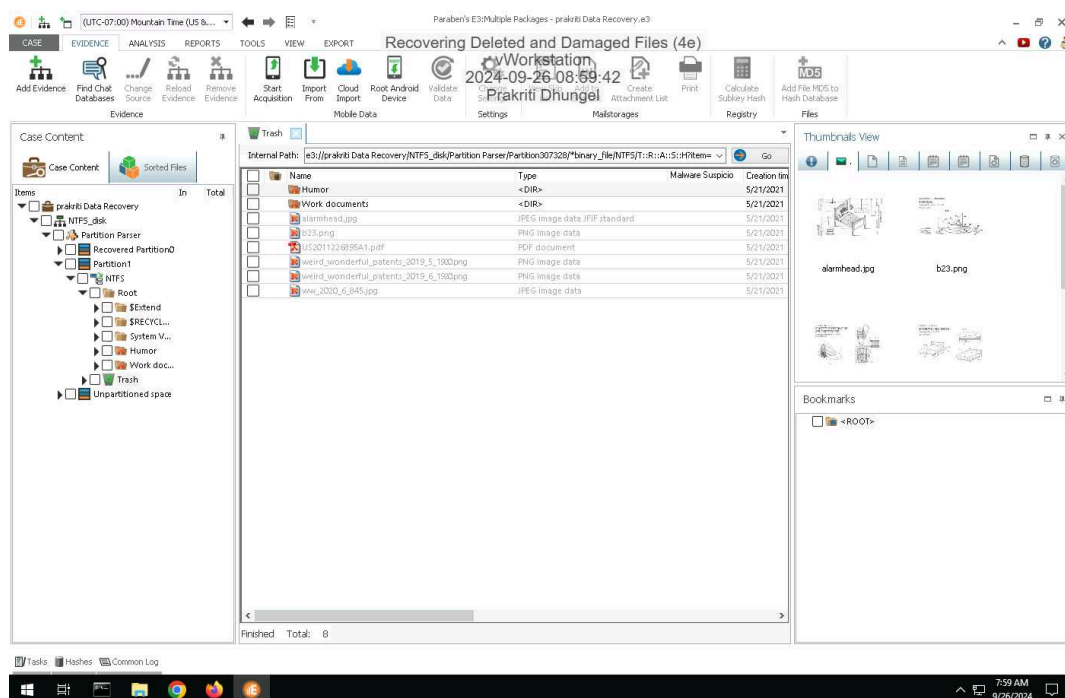
100%

Report Generated: Thursday, September 26, 2024 at 3:44 PM

Section 1: Hands-On Demonstration

Part 1: Recover Deleted Files from an NTFS Drive Image with E3

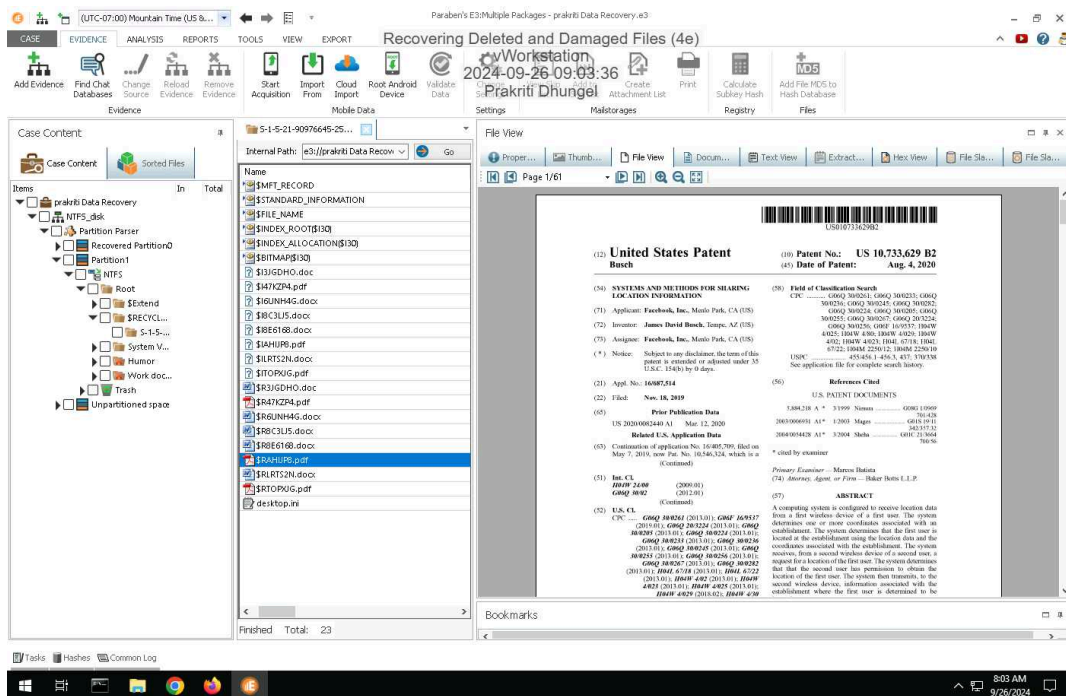
13. Make a screen capture showing the list of recovered files and folders in the E3 Trash folder.



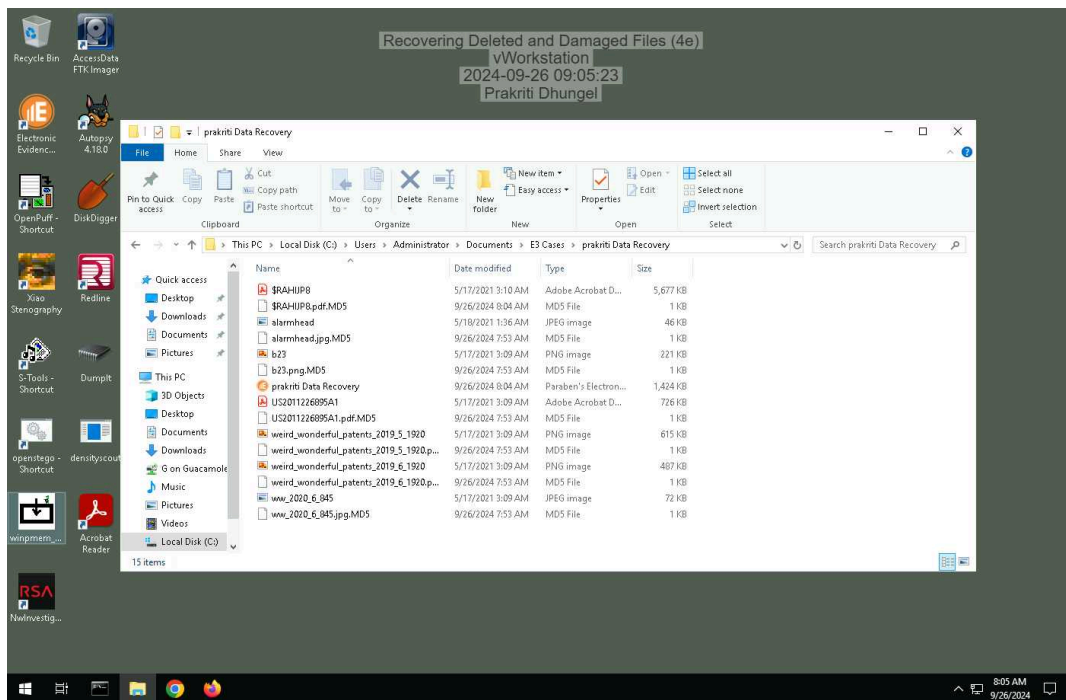
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20. Make a screen capture showing the patent file in the File Viewer.



25. Make a screen capture showing the recovered files in the File Explorer.

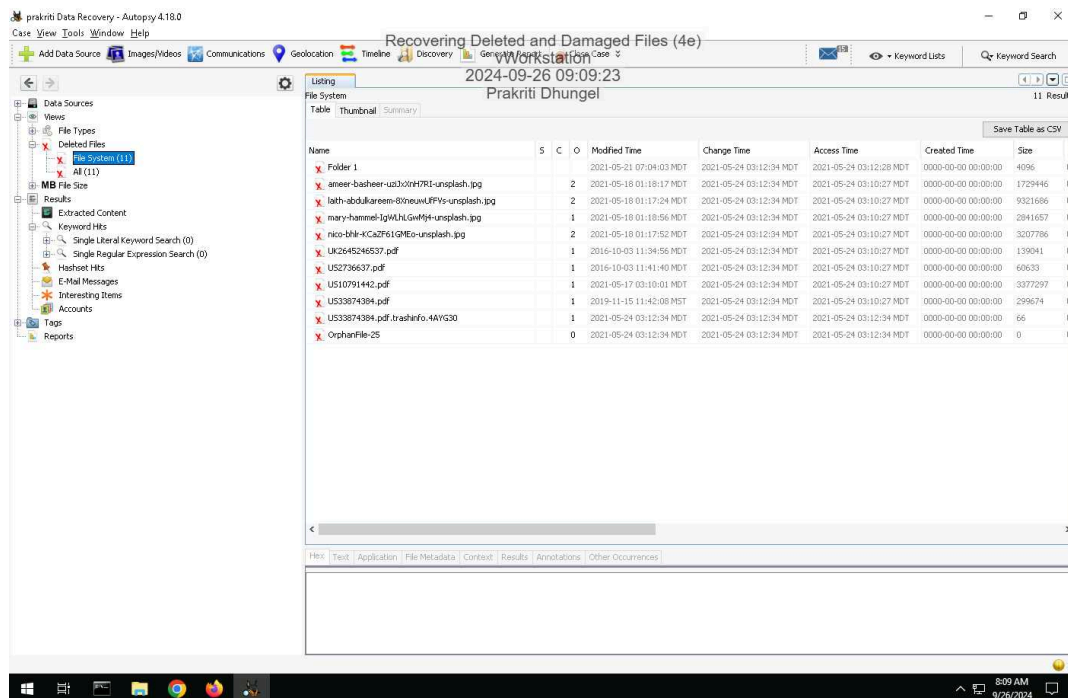


Part 2: Recover Deleted Files from an Ext4 Drive Image with Autopsy

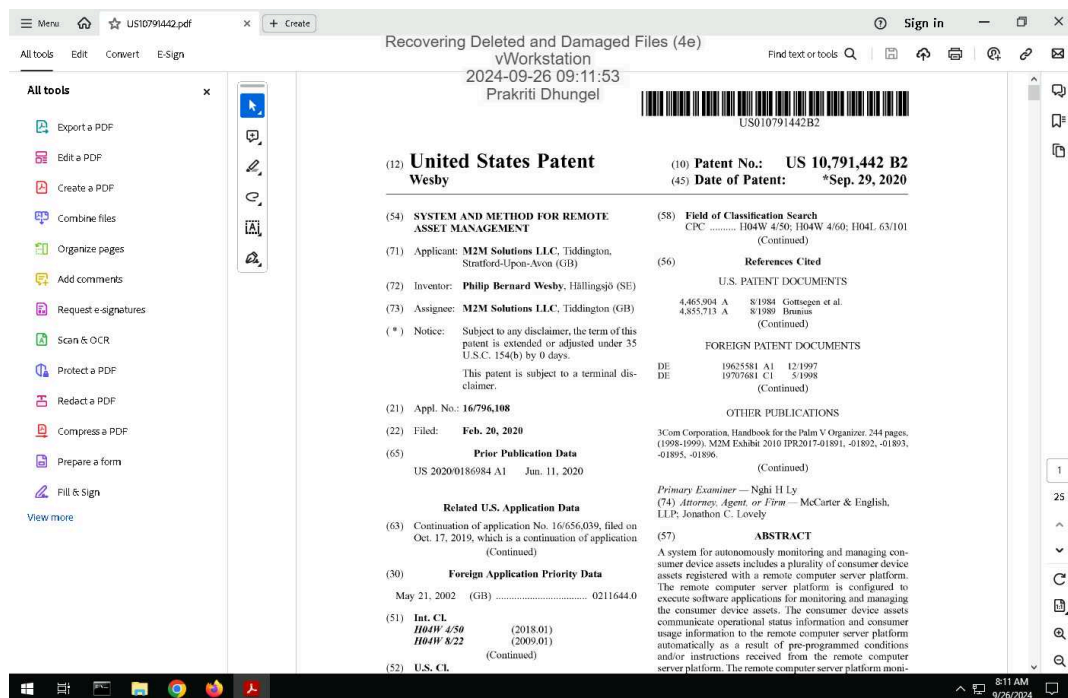
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14. Make a screen capture showing the contents of the list of deleted files in Autopsy.



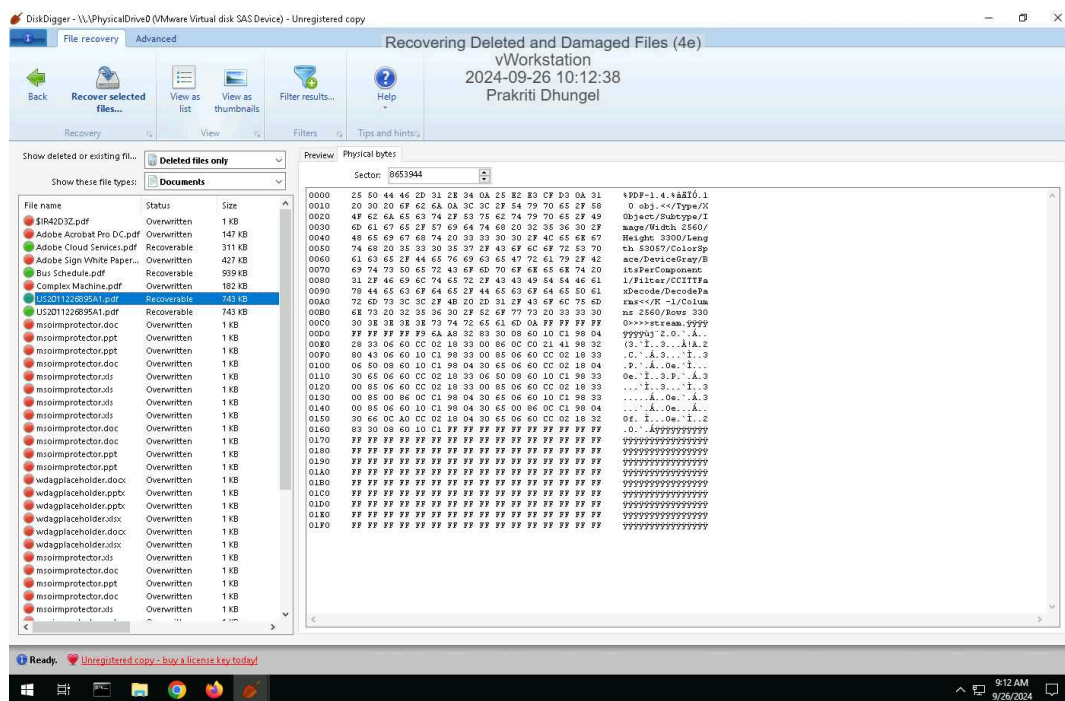
22. Make a screen capture showing the recovered patent file.



Section 2: Applied Learning

Part 1: Recover Deleted Files in Windows with DiskDigger

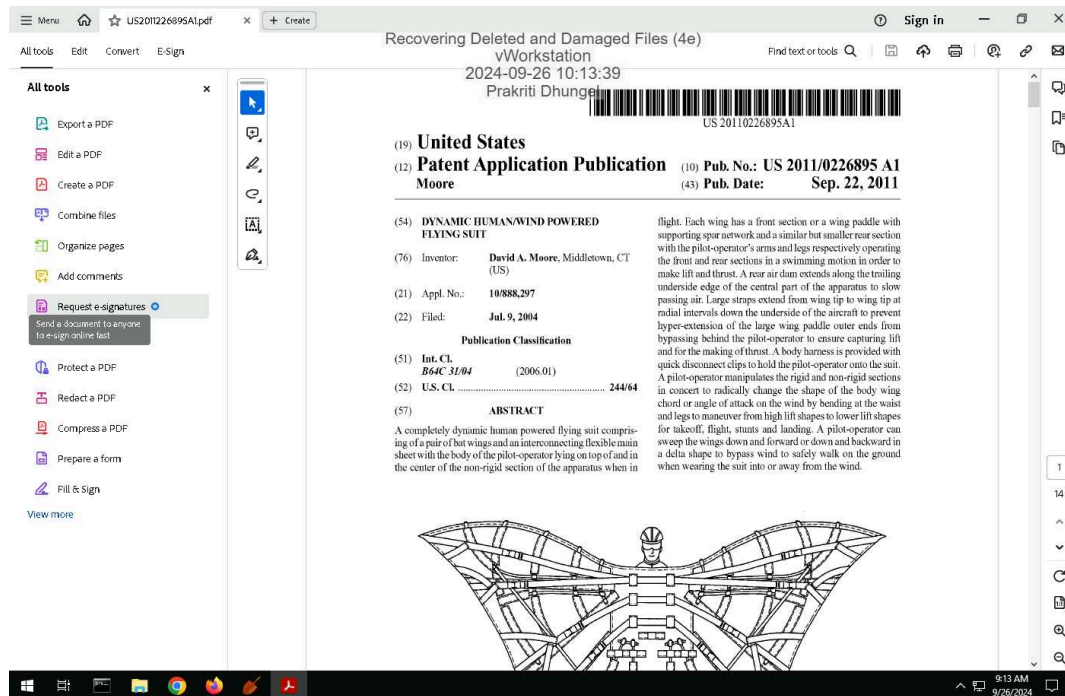
9. Make a screen capture showing the deleted patent file in DiskDigger.



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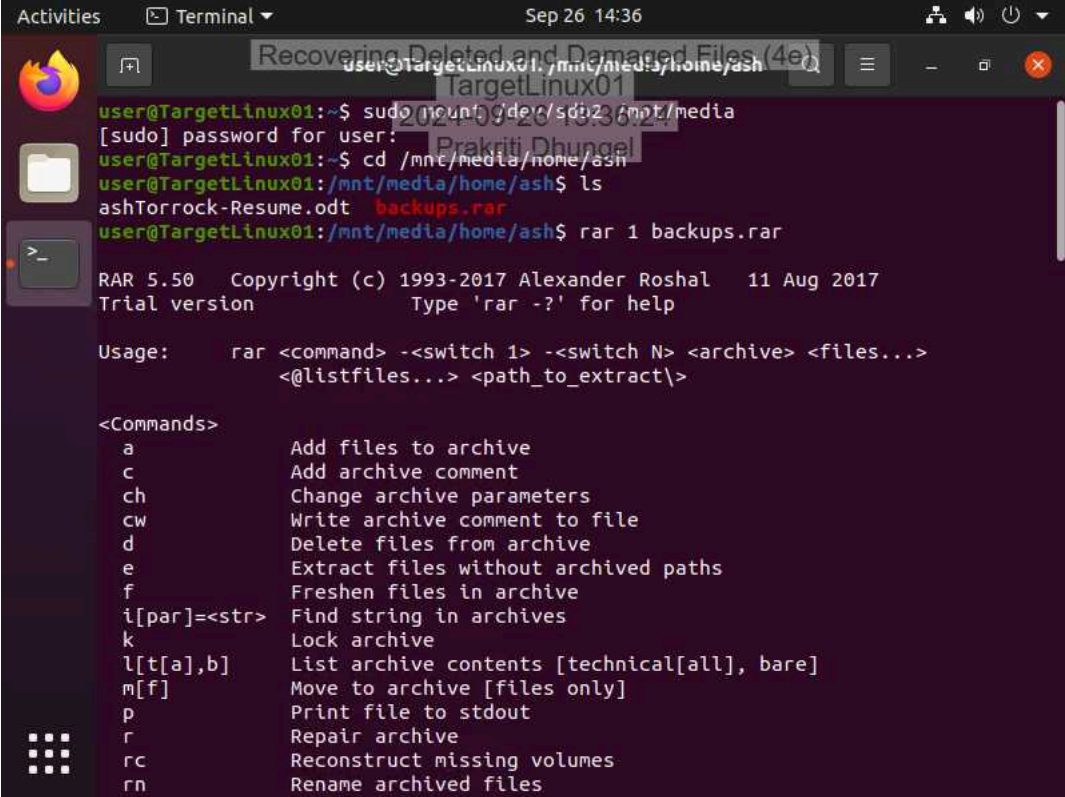
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15. Make a screen capture showing the recovered patent file.



Part 2: Recover Deleted Files in Linux with PhotoRec

9. **Make a screen capture** showing the **contents of the RAR archive in the /mnt/media/home/ash directory.**

A terminal window titled 'Terminal' with a date and time of 'Sep 26 14:36'. The window shows a series of commands and their outputs. The user is at a prompt 'user@TargetLinux01:~\$' and runs 'sudo mount -O de/ /mnt/media'. The prompt changes to '[sudo] password for user:'. The user then runs 'cd /mnt/media/home/ash' and 'ls', which shows 'ashTorrock-Resume.odt' and 'backups.rar'. The user then runs 'rar 1 backups.rar', which displays the RAR 5.50 version information and a list of commands. The terminal output is as follows:

```
user@TargetLinux01:~$ sudo mount -O de/ /mnt/media
[sudo] password for user:
user@TargetLinux01:~$ cd /mnt/media/home/ash
user@TargetLinux01:/mnt/media/home/ash$ ls
ashTorrock-Resume.odt  backups.rar
user@TargetLinux01:/mnt/media/home/ash$ rar 1 backups.rar

RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

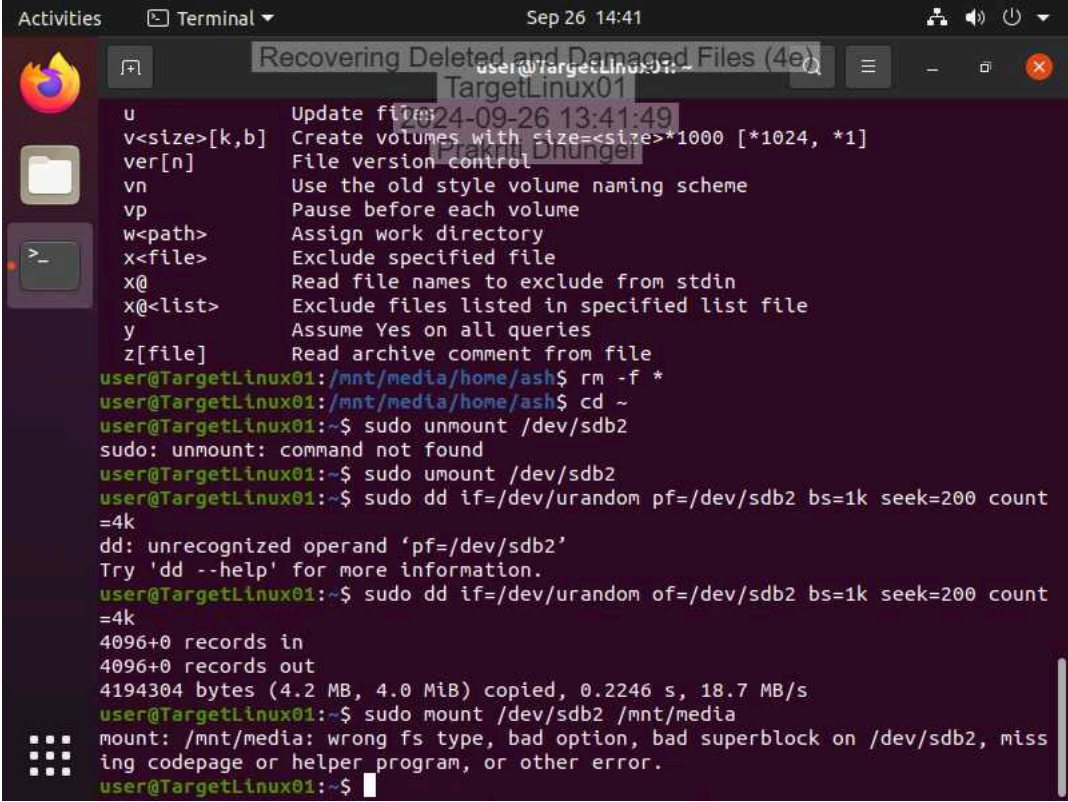
Usage: rar <command> -<switch 1> -<switch N> <archive> <files...>
        <@listfiles...> <path_to_extract\>

<Commands>
a      Add files to archive
c      Add archive comment
ch     Change archive parameters
cw     Write archive comment to file
d      Delete files from archive
e      Extract files without archived paths
f      Freshen files in archive
i[par]=<str> Find string in archives
k      Lock archive
l[t[a],b] List archive contents [technical[all], bare]
m[f]   Move to archive [files only]
p      Print file to stdout
r      Repair archive
rc     Reconstruct missing volumes
rn     Rename archived files
```


Recovering Deleted and Damaged Files (4e)

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15. Make a screen capture showing the **failed mount attempt on the /dev/sdb2 device**.

A terminal window titled "Recovering Deleted and Damaged Files (4e)" on a system named "TargetLinux01". The terminal shows a list of cfdisk options (u, v, ver, vn, vp, w, x, x@, x@<list>, y, z) and then a series of commands: 'rm -f *', 'cd ~', 'sudo umount /dev/sdb2' (which fails with "command not found"), 'sudo umount /dev/sdb2', 'sudo dd if=/dev/urandom pf=/dev/sdb2 bs=1k seek=200 count=4k' (which fails with "unrecognized operand 'pf='), and 'sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count=4k' (which succeeds, copying 4.2 MB). Finally, 'sudo mount /dev/sdb2 /mnt/media' is executed, failing with the message: "mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, missing codepage or helper program, or other error." The terminal window has a dark background and standard Linux terminal icons on the left.

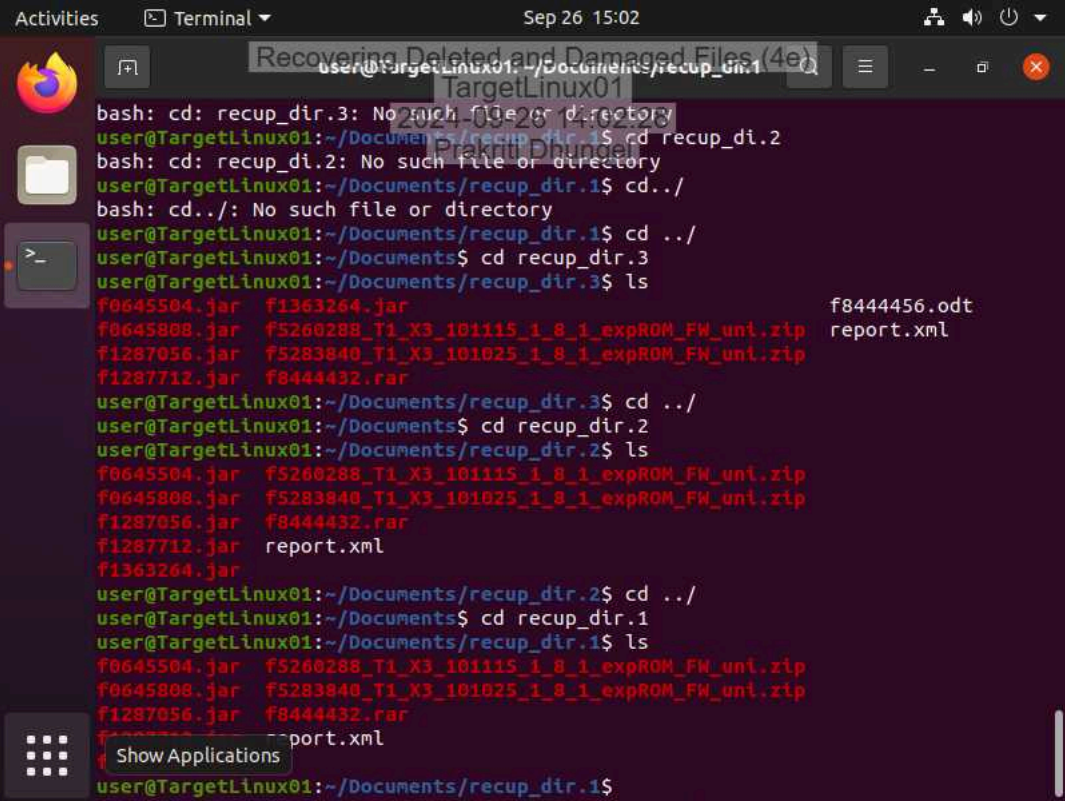
```
user@TargetLinux01:~$ cfdisk /dev/sdb
u          Update file
v<size>[k,b] Create volumes with size=<size>*1000 [*1024, *1]
ver[n]      File version control
vn         Use the old style volume naming scheme
vp         Pause before each volume
w<path>     Assign work directory
x<file>     Exclude specified file
x@         Read file names to exclude from stdin
x@<list>    Exclude files listed in specified list file
y          Assume Yes on all queries
z[file]     Read archive comment from file

user@TargetLinux01:/mnt/media/home/ash$ rm -f *
user@TargetLinux01:/mnt/media/home/ash$ cd ~
user@TargetLinux01:~$ sudo umount /dev/sdb2
sudo: umount: command not found
user@TargetLinux01:~$ sudo umount /dev/sdb2
user@TargetLinux01:~$ sudo dd if=/dev/urandom pf=/dev/sdb2 bs=1k seek=200 count
=4k
dd: unrecognized operand 'pf=/dev/sdb2'
Try 'dd --help' for more information.
user@TargetLinux01:~$ sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count
=4k
4096+0 records in
4096+0 records out
4194304 bytes (4.2 MB, 4.0 MiB) copied, 0.2246 s, 18.7 MB/s
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, miss
ing codepage or helper program, or other error.
user@TargetLinux01:~$
```

Recovering Deleted and Damaged Files (4e)

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32. Make a screen capture showing the compressed files recovered by PhotoRec.

A terminal window titled "Recovering Deleted and Damaged Files (4e)" with a subtitle "TargetLinux01" and a timestamp "2024-09-26 14:02:26". The terminal shows a series of commands and outputs for navigating through directories and listing files. The files listed include various .jar, .zip, and .rar files, as well as .odt and .xml files. The terminal output is as follows:

```
user@TargetLinux01:~/Documents/recup_dir.1$ cd recup_dir.1
bash: cd: recup_dir.3: No such file or directory
user@TargetLinux01:~/Documents/recup_dir.1$ cd recup_dir.2
bash: cd: recup_dir.2: No such file or directory
user@TargetLinux01:~/Documents/recup_dir.1$ cd ../
bash: cd ../: No such file or directory
user@TargetLinux01:~/Documents/recup_dir.1$ cd ../
user@TargetLinux01:~/Documents$ cd recup_dir.3
user@TargetLinux01:~/Documents/recup_dir.3$ ls
f0645504.jar  f1363264.jar                                f8444456.odt
f0645808.jar  f5260288_T1_X3_101115_1_8_1_expROM_FW_uni.zip  report.xml
f1287056.jar  f5283840_T1_X3_101025_1_8_1_expROM_FW_uni.zip
f1287712.jar  f8444432.rar
user@TargetLinux01:~/Documents/recup_dir.3$ cd ../
user@TargetLinux01:~/Documents$ cd recup_dir.2
user@TargetLinux01:~/Documents/recup_dir.2$ ls
f0645504.jar  f5260288_T1_X3_101115_1_8_1_expROM_FW_uni.zip
f0645808.jar  f5283840_T1_X3_101025_1_8_1_expROM_FW_uni.zip
f1287056.jar  f8444432.rar
f1287712.jar  report.xml
f1363264.jar
user@TargetLinux01:~/Documents/recup_dir.2$ cd ../
user@TargetLinux01:~/Documents$ cd recup_dir.1
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f0645504.jar  f5260288_T1_X3_101115_1_8_1_expROM_FW_uni.zip
f0645808.jar  f5283840_T1_X3_101025_1_8_1_expROM_FW_uni.zip
f1287056.jar  f8444432.rar
f1287712.jar  report.xml
Show Applications
user@TargetLinux01:~/Documents/recup_dir.1$
```


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35. **Make a screen capture** showing the **backup files recovered** from the RAR archive.

The screenshot shows a Kali Linux terminal window with the following content:

```

Activities  Terminal  Sep 26 15:03
Recovering Deleted and Damaged Files (4e)
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f0645504.jar  f5260288_T1_X3_101115_1_8_1_expROM_FW_uni.zip
f0645808.jar  f5283840_T1_X3_101025_1_8_1_expROM_FW_uni.zip
f1287056.jar  f8444432.rar
f1287712.jar  report.xml
f1363264.jar

user@TargetLinux01:~/Documents/recup_dir.2$ cd ../
user@TargetLinux01:~/Documents$ cd recup_dir.1
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f0645504.jar  f5260288_T1_X3_101115_1_8_1_expROM_FW_uni.zip
f0645808.jar  f5283840_T1_X3_101025_1_8_1_expROM_FW_uni.zip
f1287056.jar  f8444432.rar
f1287712.jar  report.xml
f1363264.jar

user@TargetLinux01:~/Documents/recup_dir.1$ sudo rar e nameofrar

RAR 5.50  Copyright (c) 1993-2017 Alexander Roshal   11 Aug 2017
Trial version                                     Type 'rar -?' for help

Cannot open nameofrar.rar
No such file or directory
No files to extract

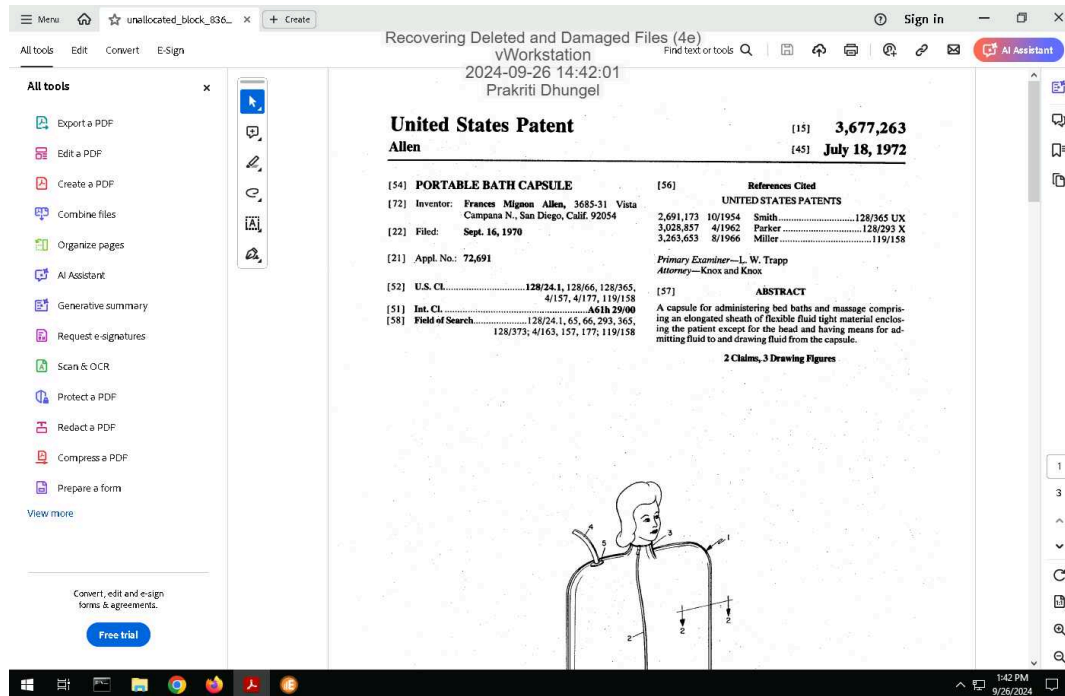
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f0645504.jar  f5260288_T1_X3_101115_1_8_1_expROM_FW_uni.zip
f0645808.jar  f5283840_T1_X3_101025_1_8_1_expROM_FW_uni.zip
f1287056.jar  f8444432.rar
f1287712.jar  report.xml
f1363264.jar

user@TargetLinux01:~/Documents/recup_dir.1$
  
```

Section 3: Challenge and Analysis

Part 1: Recover Deleted Files from a FAT Drive Image

Make a screen capture showing the patent file recovered from the FAT32 drive image within E3.



Part 2: Recover Deleted Files from a APFS Drive Image

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Make a screen capture showing the patent file recovered from the APFS drive image within Autopsy.

